APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination

MODEL QUESTION PAPER — SET 5

Course Code: PCCST501 **Course Name:** Computer Networks

Maximum Marks: 60 **Duration:** 2 hours 30 minutes

PART A — Answer all questions. Each question carries 3 marks.

Q. No.	Question	Marks	CO
1	Explain how protocol layering supports communication between applications on different hosts.	3	CO1
2	Distinguish between HTTP and electronic mail as application-layer services.	3	CO1
3	Explain the purpose of TCP sequence numbers in reliable byte-stream communication.	3	CO2
4	Describe the purpose of listen() and accept() in a TCP server program.	3	CO2
5	Explain how unicast routing supports host-to-host packet delivery.	3	CO3
6	Describe the role of Quality of Service requirements in network-layer service delivery.	3	CO3
7	Explain the role of multiple-access protocols when devices share a link.	3	CO4
8	Describe the relationship between data, signals and transmission media.	3	CO5

PART B — Answer one full question from each module. Each question carries 9 marks.

Module 1 — Question 9 (CO: CO1; Marks: 9)

(a) Explain how protocol layering separates application services from transport and network delivery functions. (4 marks)
(b) Describe the sequence of DNS resolution and HTTP communication when a browser retrieves a resource from a named website. (5 marks)

Module 1 — Question 10 (CO: CO1; Marks: 9)

(a) Compare the client-server and peer-to-peer paradigms in terms of service location and distribution of responsibilities. (4 marks)
(b) Explain how BitTorrent supports distributed file sharing through piece exchange among peers. (5 marks)

Module 2 — Question 11 (CO: CO2; Marks: 9)

(a) Explain how TCP sequence numbers and acknowledgements support reliable, ordered byte-stream delivery. (5 marks)
(b) A TCP sender transmits a segment beginning at sequence number 7200 with 450 data bytes, followed by another segment carrying 300 data bytes in order. Determine the sequence number of the first byte in the third segment. (4 marks)

Module 2 — Question 12 (CO: CO3; Marks: 9)

(a) Explain the role of network-layer routing in host-to-host delivery and distinguish intra-domain from inter-domain routing. (4 marks)
(b) A network has the following link costs: M–N = 2, M–O = 6, N–O = 2, N–P = 7, O–P = 1. Compute the minimum-cost path from M to P and its total cost. (5 marks)

Module 3 — Question 13 (CO: CO4; Marks: 9)

(a) Explain the purpose of Ethernet framing and link-layer addresses in local network communication. (5 marks)
(b) Describe how a switch uses learned address information to decide where to forward an Ethernet frame. (4 marks)

Module 3 — Question 14 (CO: CO4; Marks: 9)

(a) Explain the purpose of data-link control and multiple-access protocols in reliable sharing of a communication link. (5 marks)
(b) Distinguish between the Data-Link Provider Interface and packet-socket access using PF_PACKET. (4 marks)

Module 4 — Question 15 (CO: CO5; Marks: 9)

(a) Compare analog and digital transmission with respect to signal representation and the impact of noise. (5 marks)
(b) A digital link transmits 15,000 bits in 5 seconds. Compute its average data rate in bits per second. (4 marks)

Module 4 — Question 16 (CO: CO5; Marks: 9)

(a) Explain how bandwidth utilization and transmission-medium characteristics affect the performance of a communication system. (5 marks)
(b) A channel has a bandwidth of 6.5 kHz. Express this bandwidth in hertz and state what bandwidth represents. (4 marks)

— End of Question Paper —

B. BLUEPRINT TABLE

Q. No.	Module	Syllabus topic	CO	Bloom's level	Marks
1	1	Protocol layering	CO1	Understand	3
2	1	HTTP and electronic mail	CO1	Understand	3
3	2	TCP sequence numbers	CO2	Understand	3
4	2	TCP server socket functions	CO2	Understand	3
5	2	Unicast routing	CO3	Understand	3
6	2	Quality of Service	CO3	Understand	3
7	3	Multiple-access protocols	CO4	Understand	3
8	4	Data, signals and transmission media	CO5	Understand	3
9	1	Protocol layering; DNS and HTTP	CO1	Apply	9
10	1	Client-server, P2P and BitTorrent	CO1	Analyse	9
11	2	TCP sequence numbers and acknowledgements	CO2	Apply	9
12	2	Host-to-host routing; intra/inter-domain routing	CO3	Apply	9
13	3	Ethernet framing, link-layer addresses and switching	CO4	Apply	9
14	3	Data-link control, multiple access, DLPI and PF_PACKET	CO4	Analyse	9
15	4	Analog/digital transmission; data rate	CO5	Apply	9
16	4	Bandwidth utilization; transmission media	CO5	Apply	9

C. VERIFICATION CHECKLIST

Check	Result
Part A marks	8 × 3 = 24 marks.
Part B marks	Four answered questions × 9 = 36 marks.
Module distribution	Exactly two Part A questions and two Part B alternatives per module.
Subdivision totals	Q9–Q16 each total 9 marks and have two subdivisions.
CO coverage	CO1, CO2, CO3, CO4 and CO5 are assessed.
Problem-solving	Q11 sequence-number calculation, Q12 least-cost path, Q15 data-rate calculation.
Numerical verification	Q11: third segment starts at 7200 + 450 + 300 = 7950. Q12: M–N–O–P costs 2+2+1 = 5. Q15: 15,000/5 = 3,000 bits/s. Q16: 6.5 kHz = 6500 Hz.
No repetition	Set 5 questions are newly composed and avoid repeating the question wording/scenarios of Sets 1–4.

Official Course Outcomes

CO1: Understand the internetworking design in terms of protocol stack and the role of various application layer protocols

CO2: Illustrate the functions of the transport layer from connectionless and connection-oriented perspectives

CO3: Identify how the network layer achieves host-to-host connectivity and caters to the diverse service requirements of the host applications

CO4: Explain the nuances of the data link layer design and demonstrate the various data link link layer protocols

CO5: Describe the fundamental characteristics of the physical layer and understand how the physical layer supports the functionalities of the top layers

D. SCHEME OF VALUATION

Part A

Q1: Layering divides communication functions into layers (1); each layer offers services to the layer above (1); enables interoperable communication (1).

Q2: HTTP transfers web resources (1.5); electronic mail supports message exchange (1.5).

Q3: Sequence numbers identify byte positions (1); support ordering (1); aid acknowledgement/retransmission tracking (1).

Q4: listen() places a socket in passive listening state (1.5); accept() accepts an incoming connection and returns a connected socket (1.5).

Q5: Unicast selects a route to one destination (1); routers use forwarding information (1); supports host-to-host packet delivery (1).

Q6: Identifies QoS needs such as delay, loss or bandwidth (1); describes service requirement (1); explains influence on packet delivery (1).

Q7: Shared medium creates contention (1); multiple-access protocol coordinates transmissions (1); valid access/collision explanation (1).

Q8: Data represents information (1); signals encode data (1); transmission media carry signals (1).

Part B

Q9(a) — 4: Layering separates functions (2); identifies application, transport and network roles (2).

Q9(b) — 5: DNS lookup (2); client obtains server address (1); HTTP request (1); server response/resource delivery (1).

Q10(a) — 4: Client-server service provider and clients (2); P2P peers share service/resource roles (2).

Q10(b) — 5: File split into pieces (1); peers request pieces (1); peers upload pieces to others (1); piece availability/distribution (1); assemble file (1).

Q11(a) — 5: Sequence numbers identify byte positions (2); acknowledgements indicate next expected byte (2); retransmission/ordering contribution (1).

Q11(b) — 4: First segment ends at 7649 (1); second begins at 7650 and ends at 7949 (2); third segment begins at 7950 (1).

Q12(a) — 4: Routing supports host-to-host packet forwarding (2); intra-domain is within one domain (1); inter-domain connects routing domains (1).

Q12(b) — 5: Path M–N–O–P (1); cost 2+2+1 (2); total cost 5 (1); confirms alternatives are more costly (1).

Q13(a) — 5: Ethernet frame structure (2); source/destination link-layer addresses (2); local delivery role (1).

Q13(b) — 4: Learns source address/port (1); looks up destination (1); forwards to matching port (1); floods unknown destination if needed (1).

Q14(a) — 5: Data-link control structures/manages link transfer (2); multiple-access coordinates shared medium (2); valid reliability/efficiency aspect (1).

Q14(b) — 4: DLPI exposes data-link services to users (2); PF_PACKET provides packet-level socket access (2).

Q15(a) — 5: Analog continuously varying signal (2); digital discrete levels (2); noise-effect comparison (1).

Q15(b) — 4: Rate = bits/time (1); 15,000/5 (2); 3,000 bits/s (1).

Q16(a) — 5: Bandwidth/capacity relationship (2); medium attenuation/noise characteristics (2); performance consequence (1).

Q16(b) — 4: 1 kHz = 1000 Hz (2); 6.5 kHz = 6500 Hz (1); bandwidth is the frequency range occupied/available for a signal (1).